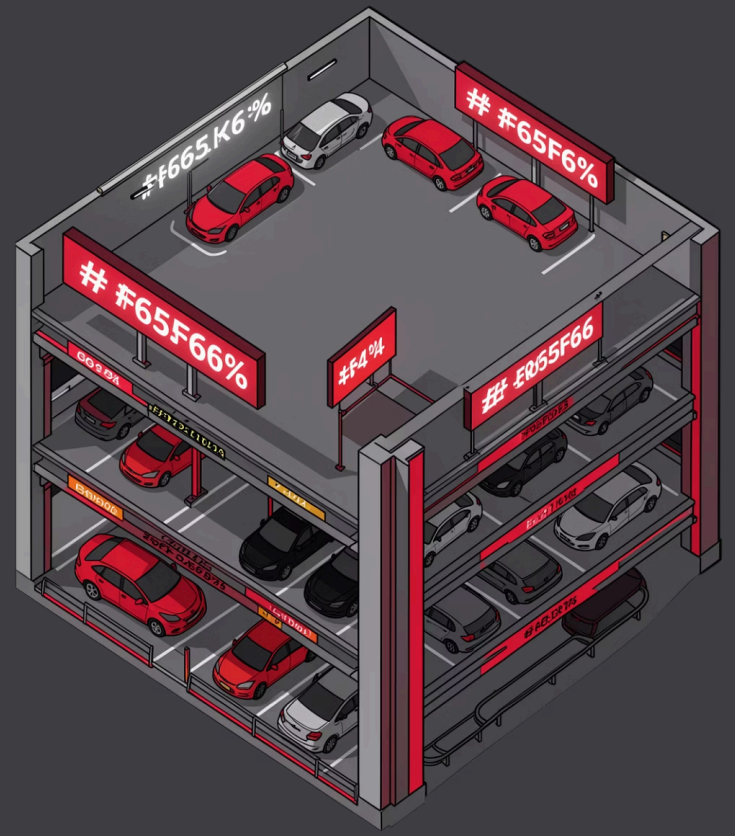
Smart Parking Lot Allocation and Billing System

A DBMS-driven solution for real-time slot allocation, reservations, and automated billing

Team: [Add team member names] · DBMS Project-Based Learning, Woxsen University



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The Problem: Manual Chaos in Parking

Conflicts & Double-Bookings

Multi-level facilities allocate slots manually — causing conflicts, wasted time, and frustrated customers.

Zero Live Visibility

Customers can't find available slots; operators can't track real-time occupancy across floors or zones.

Inconsistent Billing

Manual calculations introduce errors and revenue leaks — no audit trail, no accountability.

No Operational Insights

Operators can't identify peak hours, revenue trends, or bottlenecks without a data layer.



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Our Solution: Database-Driven Parking Intelligence



Real-Time Slot Search

Find and allocate slots instantly by vehicle type across floors and zones.



Reservation Management

Book slots with automatic expiry handling for unclaimed reservations.



Automated Billing

Entry/exit logs drive duration-based billing — no manual calculation needed.



Pass Management

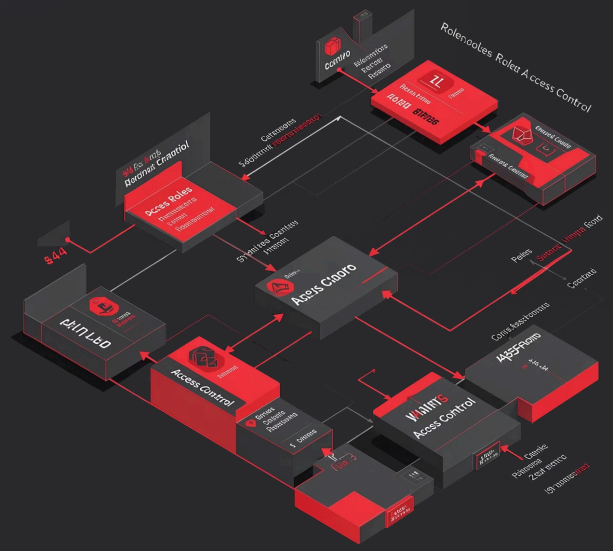
Monthly, quarterly, and annual passes issued and tracked in the database.



Operator Reporting

Live occupancy, revenue summaries, and peak-hour insights for decision-makers.

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System Scope & User Roles

Scope: Single facility with multiple floors, zones, and slots; supports four vehicle types — two-wheeler, car, SUV, and truck.

👤 Customer

Reserve slots, park vehicles, make payments

🏁 Gate Operator

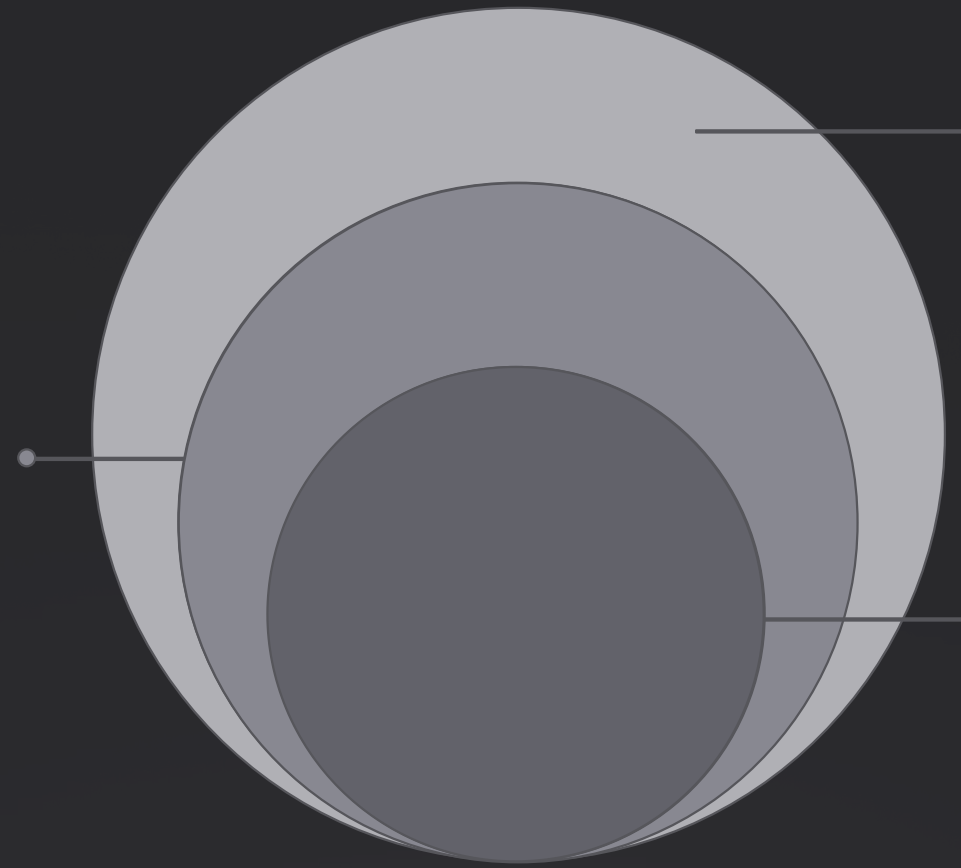
Check-in/check-out vehicles, verify reservations

🛠️ Admin

View reports, manage tariffs, monitor revenue

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Three-Tier System Architecture



Storage Layer

Persistent storage with backup and recovery

Database Layer

SQLite/MySQL enforces integrity via constraints

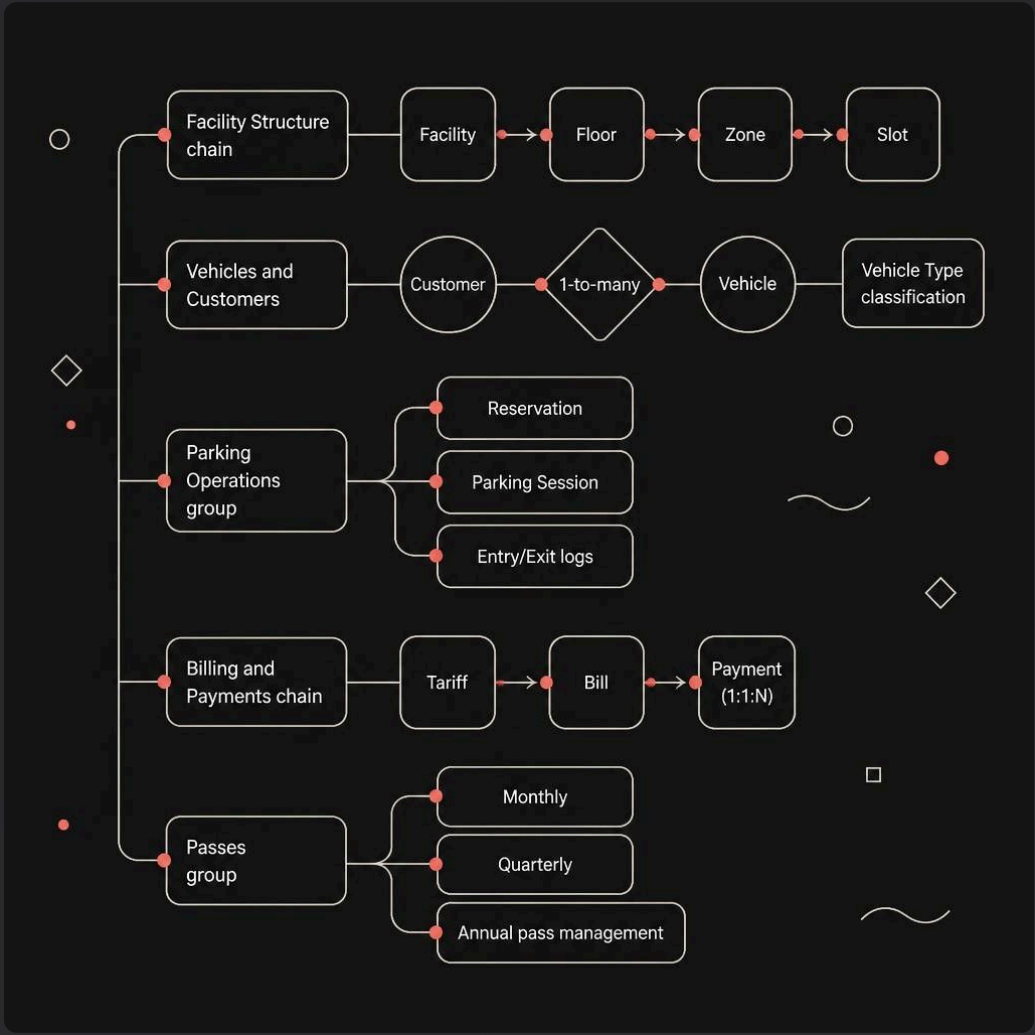
Application Layer

Python CLI handles input and business logic

Separation of concerns across all three layers ensures long-term scalability and maintainability of the system.

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Entity-Relationship Model: 13 Core Entities



Entity Groups

Facility Structure

Facility → Floor → Zone → Slot (hierarchical 1:N chain)

Vehicles & Customers

Customer → Vehicle (1:N), classified by Vehicle Type

Parking Operations

Reservation, Parking Session, Entry/Exit logs

Billing & Payments

Tariff → Bill → Payment (1:1:N chain)

Passes

Monthly, Quarterly, and Annual pass management

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Relational Schema: Normalized to 3NF

Design Principles

1 No Partial Dependencies

All 13 tables eliminate partial dependencies — every non-key attribute depends on the whole primary key.

2 No Transitive Dependencies

Non-key attributes depend only on the primary key, never on another non-key attribute.



3 Referential Integrity

Foreign keys enforce relationships across all tables — no orphaned records.

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Business Rules Enforced at Database Layer

One Active Vehicle Per Slot

Trigger prevents double-booking — no two sessions can occupy the same slot simultaneously.

Vehicle Type Matching

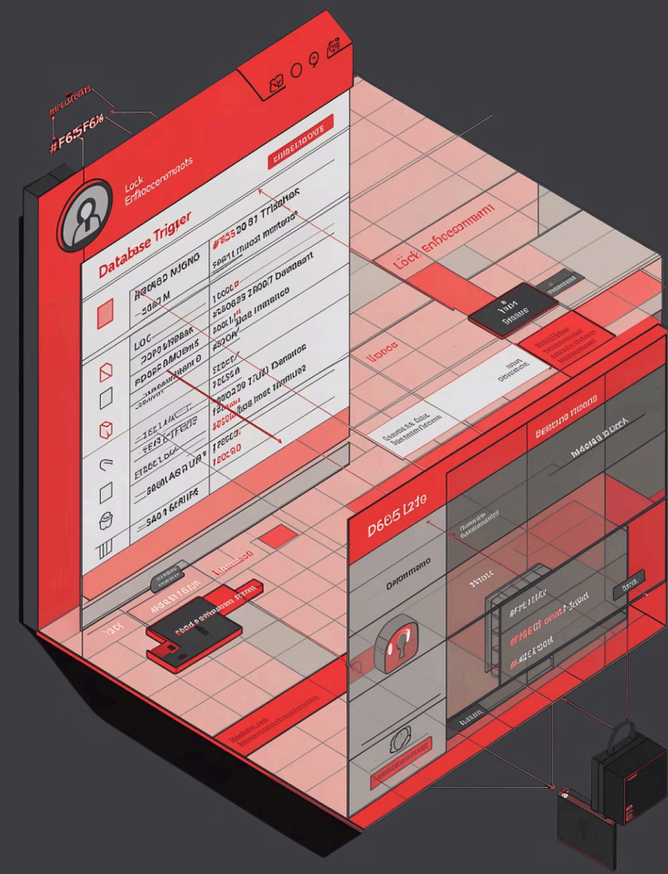
Trigger validates that the vehicle type matches the slot's designated classification before entry.

Temporal Integrity

CHECK constraint ensures exit time is always greater than entry time — no logical inconsistencies.

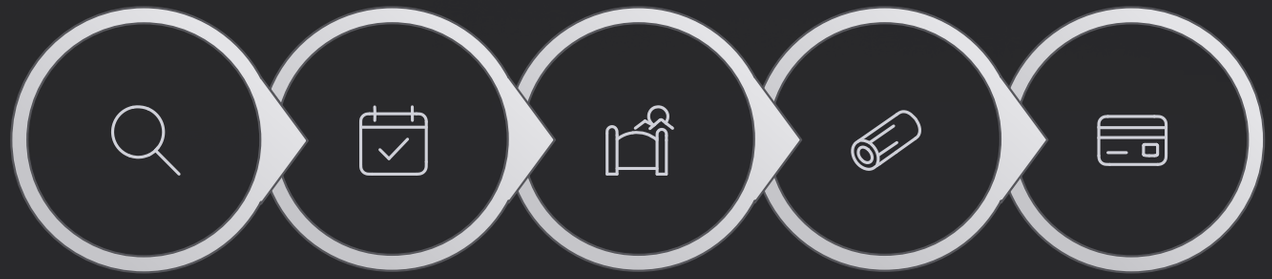
Billing Formula

Amount = (duration × hourly rate), capped at daily maximum — calculated and stored automatically.



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Core CRUD Operations



Operation Summary

Search

Find free slots filtered by vehicle type and location preference

Reserve & Entry

Book a slot, then gate check-in validates the reservation on arrival

Exit & Billing

Departure is logged and a bill is auto-generated from session duration

Payments & Passes

Payments recorded against bills; multi-duration passes issued and tracked

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Sample SQL Queries: Joins, Aggregates & Views

Query Type	Use Case	Example
Joins	Currently parked vehicles with customer details	Session □ Vehicle □ Customer — returns name, plate, duration
Aggregates	Revenue by day, avg. session duration, occupancy rate	SUM(amount) GROUP BY date, AVG(duration) per vehicle type
Subqueries	Reservations expired without a corresponding session	SELECT … WHERE NOT EXISTS (SELECT 1 FROM Session WHERE …)
Views	Live occupancy by zone, revenue by vehicle type	CREATE VIEW v_occupancy AS SELECT zone, COUNT(*) FROM Slot …

All query types are implemented and tested against the live SQLite/MySQL schema with real session data.

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